AC CHEMISTRY · UNIT 1 · MEASUREMENT & DENSITY

Accurate, Precise, Both, Neither

Two independent faults in one balance, and which of them repeating a measurement can actually remove.

Name ______________________ Date ______________________ Period ______________________

Demonstration: andresdbp.github.io/resources/demos/accurate-precise/

1 Before you open anything

Commit to an answer on paper first. The demonstration will settle it exactly, so it is worth being specific rather than safe.

PREDICTION

A balance is out of calibration and reads **0.30 g high**. It is otherwise superb: repeat weighings of the same object agree to about ±0.02 g. You place a certified **10.000 g** standard on it, weigh it **500 times**, and average all 500 readings.

Write the value you expect that average to have, to three decimal places, and how far you expect it to sit from 10.000 g.

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In one or two sentences, say what reasoning got you that number.

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2 The four cases, on one run

Open the demonstration. The two sliders are *Systematic bias* and *Random scatter (σ)*, and they act on the same set of weighings — moving a slider reshapes the run you already have rather than starting a new one. That lets you compare all four cases on identical data.

1. Press **New run**, then press **Add 25** four times so that **Trials n** reads 100.
2. Click the preset button **Accurate and precise**. Do not press any trial button again — n must stay at 100 for every row.
3. Fill in the first row from the readouts and from the heading in the coloured verdict panel.
4. Click each of the other three presets in turn and fill in the remaining rows.

Bias	σ	Verdict panel heading	Mean	Mean - true	Reading SD
0.00 g	0.02 g				
+0.30 g	0.02 g				
0.00 g	0.15 g				
+0.30 g	0.15 g				

a. Which readout tracks the *bias* slider, and which tracks the *scatter* slider? Name one readout for each.

b. Rows 1 and 2 have the same Reading SD but different Mean - true. Rows 1 and 3 have the same Mean - true but different Reading SD. State in one sentence what that pattern shows about the two faults.

> The demonstration calls a balance **accurate** when |bias| ≤ 0.05 g and **precise** when σ ≤ 0.05 g. Those cut-offs are a choice made for this demonstration, not a law of nature — a real tolerance is set by what the measurement is for.

3 What more trials buy you

Here you vary one thing only: the number of weighings. Both sliders stay put.

1. Click the preset **Neither** (bias +0.30 g, σ 0.15 g), then press **New run** to clear the trials. The sliders keep their settings.
2. Press **Add 25** once. Record the n = 25 row.
3. Press it again for n = 50, twice more for n = 100, four more for n = 200, eight more for n = 400.

Trials n	Std. error	Predicted SE	Mean - true
25			
50			
100			
200			
400			

a. From n = 25 to n = 100 the number of trials rose by a factor of 4. By what factor did *Predicted SE* fall? Check the same factor from n = 100 to n = 400.

b. Complete the rule in your own words: to cut the standard error in half you must multiply the number of trials by ____, because standard error falls as ____________.

c. Over that same 16-fold increase in trials, what happened to *Mean - true*? Quote your first and last values.

d. Go back to your prediction in Part 1. Compare it with what you have just recorded. If it does not match, say precisely which step of your reasoning the data contradicts.

4 Working backwards from a requirement

A question rarely hands you n and asks for the error. It hands you an error you must meet and asks how to get there.

a. With σ = 0.15 g, how many weighings are needed to bring the standard error down to 0.010 g? Work it out on paper first, showing the rearrangement.

1. Press **New run**. Leave the sliders at bias +0.30 g and σ 0.15 g.
2. Press **Add 25** until **Trials n** reaches the number you calculated. Read **Predicted SE**.

b. Predicted SE at that n: _______________ g. Did you hit 0.010 g? If you missed slightly, note that the demonstration also folds in the rounding of the display to 0.01 g — explain which direction that pushes the value.

c. Now press **New run**, set **Systematic bias** to **-0.17 g** and **Random scatter (σ)** to **0.05 g**. *Before* pressing anything else, predict the *Mean* and *Mean - true* readouts after 100 weighings.

Now press **Add 25** four times. Actual Mean: _______________ g. Actual Mean - true: _______________ g.

d. Your prediction in (c) will not match the last digit exactly. Compare the size of that disagreement with the *Std. error* readout. Are they the same order of magnitude, and why should they be?

e. The shortcut “our readings scatter, so we will take more of them and the average will be accurate” is doing real work in a lot of lab reports. Keeping the settings from (c), press **Run to 500**. State exactly when that shortcut holds and when it does not, referring to both sliders.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it and check yourself — and where you were wrong, write down which step went off.

1. A certified 25.000 g standard is weighed five times on the same balance: 25.12, 25.11, 25.13, 25.12, 25.12 g. Find the mean, the bias, the sample standard deviation and the standard error. Using the ±0.05 g cut-offs from Part 2, classify the balance as accurate, precise, both or neither.

2. A balance has $\sigma = 0.12$ g and negligible bias. How many weighings are needed for the standard error to fall below 0.015 g? Give the smallest whole number that works.

3. Two balances are each used four times on the same certified 10.000 g standard.
Balance A: 10.31, 10.29, 10.30, 10.30 g.
Balance B: 9.8, 10.2, 10.1, 9.9 g.
Compute the mean and standard deviation for each. Which balance would you rather use for a density determination, and why? Your answer must say what you would do with the balance you chose.

4. Explain, without computing anything, why weighing your *unknown* two hundred times can never reveal that the balance is out of calibration, and name one measurement that would. Your explanation should refer to what averaging does to each of the two error types.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Drag *Systematic bias* to -0.40 g and watch which side of the green line the red dashed line sits on. Does the width of the 95 % band change at all when you move the bias slider? Should it?
- Set σ to 0.00 g and then to 0.01 g with n around 100. Watch *Reading SD*, *Std. error* and *Predicted SE*. The model adds a rounding term because the display only shows 0.01 g — see whether you can spot it in the numbers.
- Fix both sliders, then press *New run* half a dozen times, stopping at n = 25 each time and noting *Mean - true*. Do the same at n = 500. How much does the answer wander in each case, and does that match the Std. error readout?
- Find a combination of the two sliders that pushes readings off the edge of the target plot, so the little arrows appear at the boundary. What does that tell you about how far out a single reading can land?